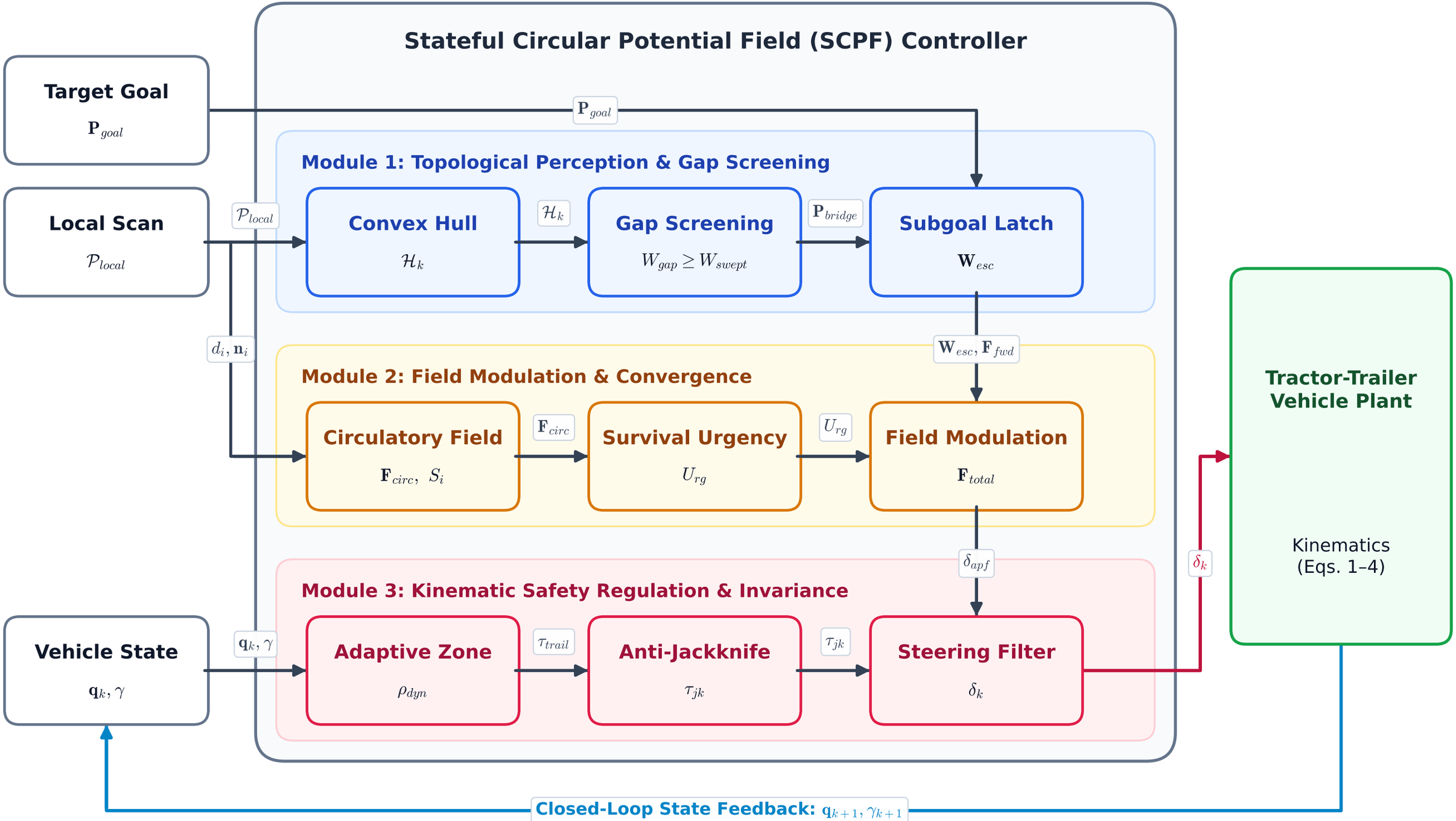
Stateful Circular Potential Field (SCPF) Controller
Target Goal
$\mathbf{P}_{goal}$
Local Scan
$\mathcal{P}_{local}$
$\mathbf{P}_{goal}$
$\mathcal{P}_{local}$
Module 1: Topological Perception & Gap Screening
Convex Hull
$\mathcal{H}_k$
$\mathcal{H}_k$
Gap Screening
$W_{gap} \geq W_{swept}$
$\mathbf{P}_{bridge}$
Subgoal Latch
$\mathbf{W}_{esc}$
$d_i, \mathbf{n}_i$
$\mathbf{W}_{esc}, \mathbf{F}_{fwd}$
Module 2: Field Modulation & Convergence
Circulatory Field
$\mathbf{F}_{circ},\ S_i$
$\mathbf{F}_{circ}$
Survival Urgency
U_{rg}
U_{rg}
Field Modulation
$\mathbf{F}_{total}$
δ_{apf}
δ_k
Module 3: Kinematic Safety Regulation & Invariance
Vehicle State
$\mathbf{q}_k, \gamma$
$\mathbf{q}_k, \gamma$
Adaptive Zone
ρ_{dyn}
τ_{trail}
Anti-Jackknife
τ_{jk}
τ_{jk}
Steering Filter
δ_k
Tractor-Trailer Vehicle Plant
Kinematics (Eqs. 1–4)
Closed-Loop State Feedback: $\mathbf{q}_{k+1}, \gamma_{k+1}$